



Customer Shopping Behavior Analysis

3,900 purchases — 18 features — goal: uncover spending patterns, segments, product & subscription insights

Dataset Snapshot

Size

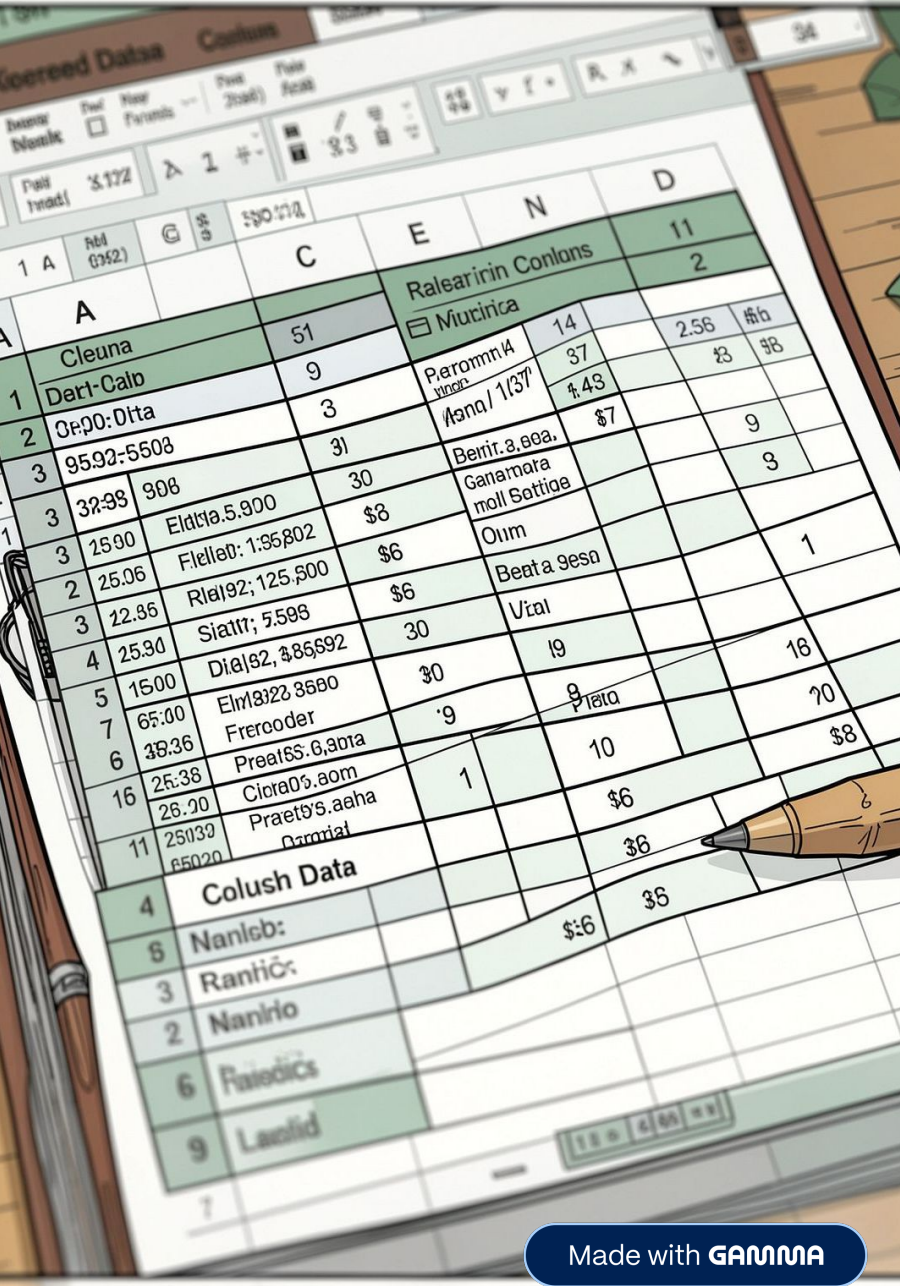
Rows: 3,900 · Columns: 18

Key Features

Demographics, purchase details,
behavior, shipping, discounts

Missing Data

Review Rating: 37 nulls



The illustration shows a spiral-bound notebook with a pencil resting on it. The notebook contains a spreadsheet with various data entries. The spreadsheet has columns labeled A, B, C, E, N, and D. The data includes names, prices, and other numerical values. Some cells are highlighted in green, and there are some handwritten notes and corrections. The overall style is a detailed, hand-drawn illustration.

	A	B	C	E	N	D
1	Cleuna	51				11
2	Dart-Calo	9				2
3	0e90:Dtta	3				
3	9592:5508	31				
3	32:98	908	30			
3	2590	Eldtla.5.900	\$8			
2	25.06	Filled: 1:35802	\$6			
3	22.36	Rlat92, 125,500	\$6			
4	25.30	Siattr, 5.598	30			
5	1500	Dlat92, 486592	\$0			
7	65:00	Elm8322 3680	\$9			
6	38.36	Frecoder				
16	25:36	Preat65.6.301a	1			
11	26:20	Cicra0's.aom				
	25032	Praet's.saha				
	65020	Camial				
4	Colush Data					
5	Nanleb:					
3	Ranlic:					
2	Nanrio					
6	Paedics					
9	Laolid					

Python – EDA & Preparation

- Loaded with pandas
- df.info() & .describe()
- Imputed review ratings by category median
- Renamed columns to snake_case
- Engineered age_group & purchase_frequency_days
- Dropped redundant promo_code_used
- Loaded cleaned DataFrame into PostgreSQL



SQL Analysis — Revenue & Segments

Revenue by Gender

Male: 157,890 • Female: 75,191

Subscribers vs Non

Yes: 1,053 (avg \$59.49, revenue 62,645) • No: 2,847 (avg \$59.87, revenue 170,436)

Customer Segments

Loyal: 3,116 • Returning: 701 • New: 83



Top Products & Ratings



Top by Rating

Gloves 3.86 • Sandals 3.84 •
Boots 3.82 • Hat 3.80 • Skirt
3.78



Top by Orders

Clothing: Blouse, Pants, Shirt •
Accessories: Jewelry,
Sunglasses, Belt

Discounts & High-Spenders



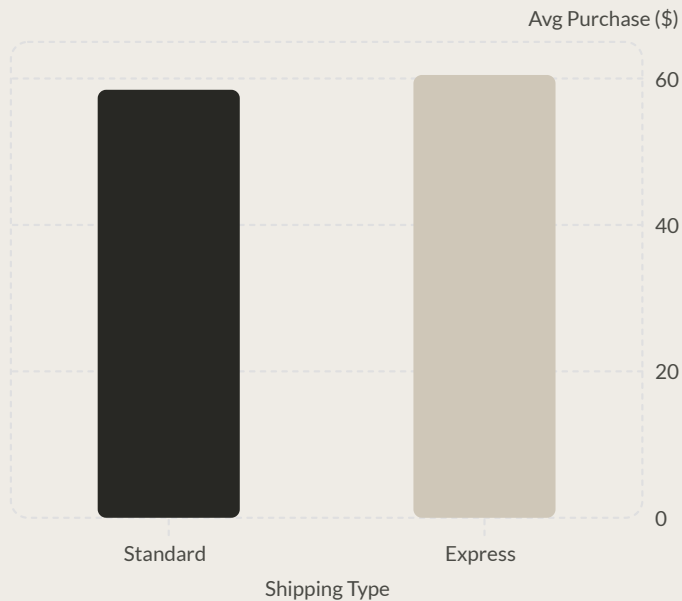
- 839 rows: customers who used discounts but spent above avg
- Top discount-dependent items: Hat 50% · Sneakers 49.66% · Coat 49.07% · Sweater 48.17% · Pants 47.37

Shipping & Purchase Behavior

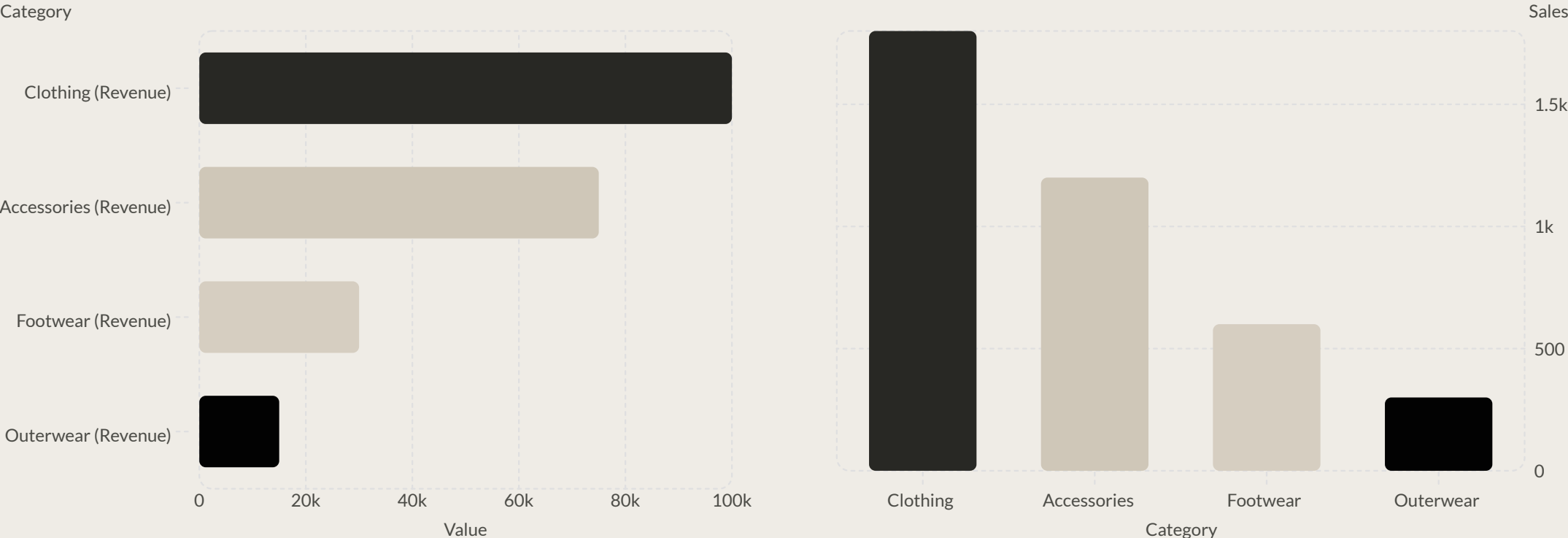
Payment: PayPal most common

Frequency mode: Every 3 Months

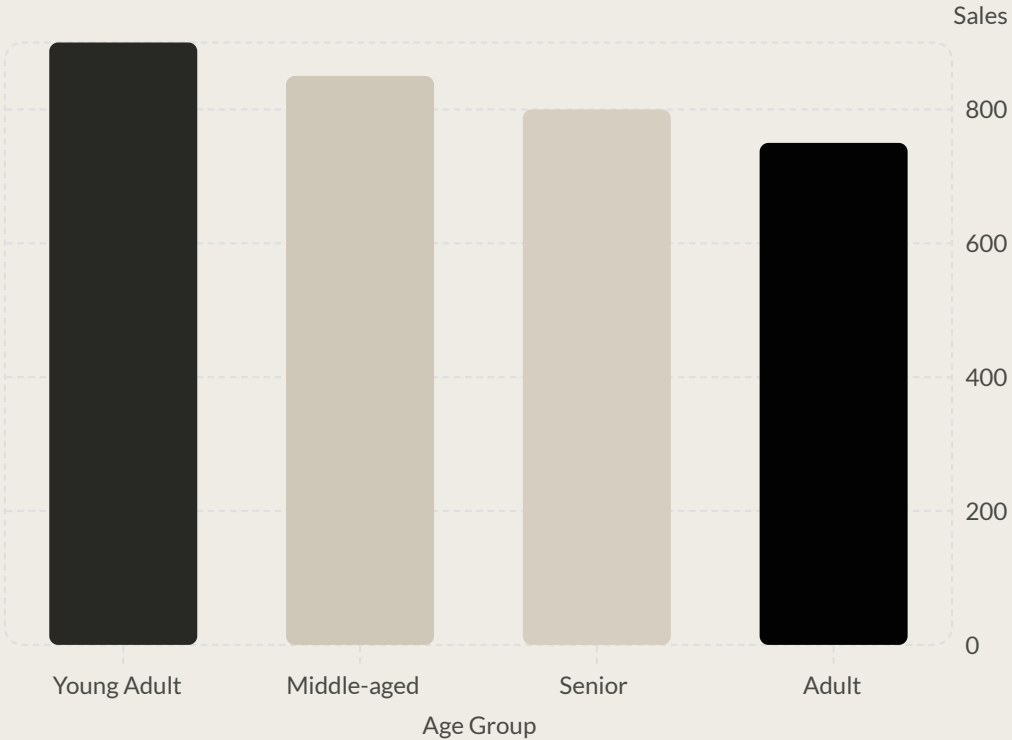
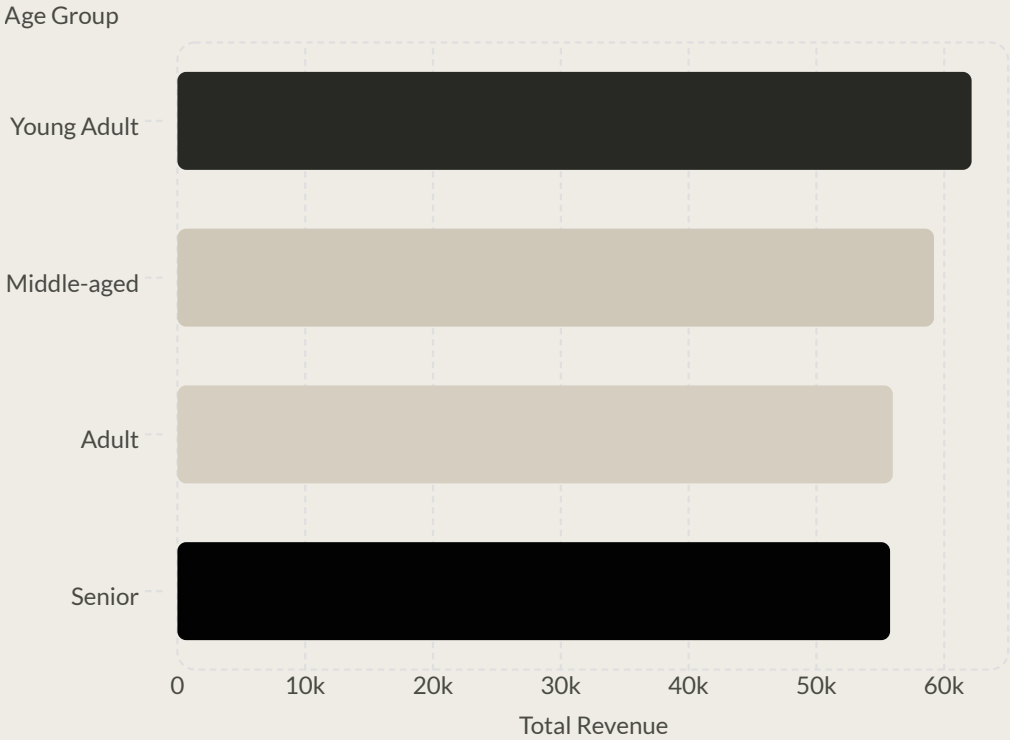
Discount_applied present; promo_code_used dropped



Revenue & Sales by Category



Age Group Revenue & Sales



Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers to grow Loyal segment

Review Discounts

Balance sales lift with margin control

Product Positioning

Feature top-rated & best-selling items in campaigns

Targeted Marketing

Focus on high-revenue age groups & express-shipping users



Dashboard built in Power BI: 3.9K customers • Avg purchase \$59.76 • Avg rating 3.75 • Subscribers 27%